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Abstract Title:	Harnessing Algorithmic Disagreement for Automated Lumbosacral Transitional Vertebrae Detection
Abstract Body:	INTRODUCTION. Using a reverse translational approach, we leveraged a surgical complication—wrong-level spine surgery due to lumbosacral transitional vertebrae (LSTV)—to drive research. LSTV affects up to 35% of patients, yet most AI segmenters force fixed L1–L5 labels onto anomalous anatomy. We propose an ensemble strategy treating spatial labeling disagreement between independent models (TotalSpineSeg [TSS] and SPINEPS) as a biomarker for LSTV. METHODS. We analyzed 282 lumbar MRIs from the RSNA 2024 Lumbar Spine challenge. TSS functions as a strict template, always assigning five lumbar labels (L1–L5), while SPINEPS acts as an anatomical sensor, labeling segments by local morphology and often detecting a transitional L6. To detect labeling failures, we developed a sequence-level alignment algorithm that maximizes 3D volumetric overlap between model outputs while "sliding" label sets across integer offsets (–2 to +2) to find the best fit. By anchoring consistent cranial levels (L1–L3), discordance is isolated to the lumbosacral junction. Disagreement regarding the lowest mobile segment (e.g., TSS labeling L5 vs. SPINEPS labeling

transitional L6) triggers automated morphometry: calculating transverse process (TP) height, TP-sacral distance, and axial signal intensity to distinguish pseudo-articulation from fusion. RESULTS. Perfect alignment (offset=0) occurred in 232 studies (82.3%). True sequence shifts of +1 (lumbarization) occurred in 19 (6.7%) and –1 (sacralization) in 24 (8.5%). Of 30 cases where SPINEPS identified L6, 21 (67.7%) were confirmed shifts; 9 (29.0%) were rejected as false positives via cranial anchoring. Mean alignment Dice was 0.857±0.028. Overall, LSTV was detected in 111 studies (39.2%). CONCLUSIONS. Treating algorithmic disagreement as signal rather than error creates a failsafe for LSTV detection. This framework enables systematic morphometric cataloging and provides surgeons a critical preoperative safeguard against wrong-level surgery.

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